A phylogenetic approach to refine taxonomic assignments of eDNA metabarcoding sequences

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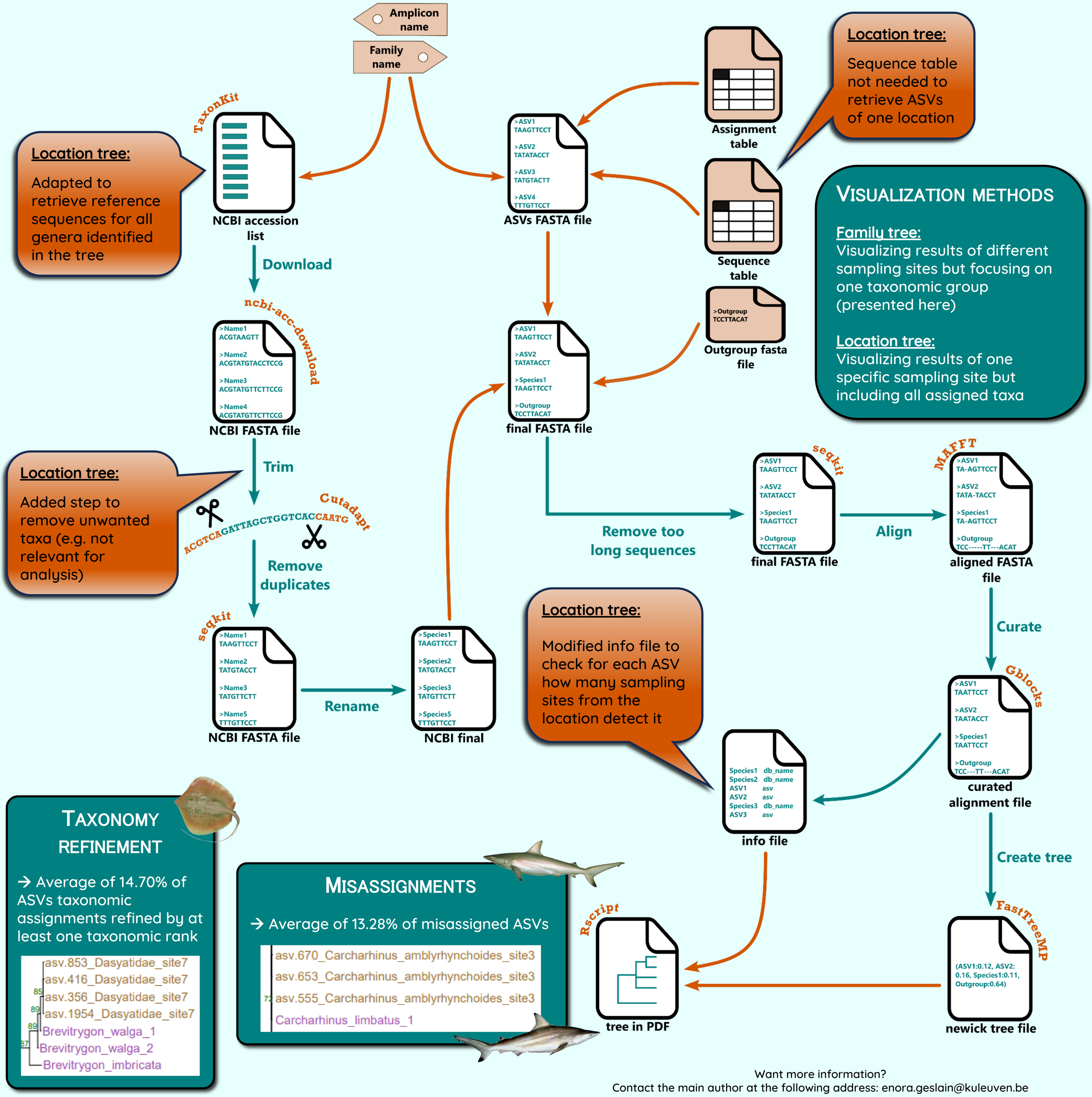
INTRODUCTION

Methods to characterize environmental DNA (eDNA) such as metabarcoding are increasingly being employed for monitoring biodiversity from genetic traces present in soil, water or even air samples. However, the bioinformatic tools based on sequence similarity used for species assignment in large scale metabarcoding analyses have limitations. Sometimes, they misassign species due to high sequence similarity between related species or fail to achieve taxonomic assignment at low taxonomic levels due to sequence resolution limitations. Here we present a tool to produce phylogenetic trees to detect problematic assignments.



VISUALIZATION METHODS

Family tree:
Visualizing results of different sampling sites but focusing on one taxonomic group (presented here)

Location tree:
Visualizing results of one specific sampling site but including all assigned taxa

TAXONOMY REFINEMENT

→ Average of 14.70% of ASVs taxonomic assignments refined by at least one taxonomic rank

MISASSIGNMENTS

→ Average of 13.28% of misassigned ASVs

Want more information?
Contact the main author at the following address: enora.geslain@kuleuven.be

EMBRC EUROPEAN MARINE BIOLOGICAL RESOURCE CENTRE

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